Description of the laboratory work No. 6

I2C PROGRAMMING OF 32-BIT MICROCONTROLLERS (STM32F429I-DISC1)

1. Objectives
2. Preparation for the work
3. Workflow

3.1. Work with a teacher

3.1.1. STM32CubeIDE

3.1.2. Template of the code

3.1.3. Main functions

3.2. Work by individual task

3.3. Literature

1. Objectives

Get to know the architecture, functionality and configuration features of the I2C serial interface on the STM32F429I-DISC1 board. Develop the ability to configure one board as I2C master and the second board as I2C slave, to exchange 1-byte and 3-byte frames in polling mode, and to display the communication results on the onboard TFT screen.

2. Preparation for the work

We will work with the I2C3 serial interface of the STM32F429I-DISC1 board during this laboratory work. Two boards are required: one board acts as the master and the second board acts

as the slave. Connect PA8 to PA8 for I2C3_SCL, PC9 to PC9 for I2C3_SDA, and GND to GND. Because this board has only two user LEDs, the transferred and received I2C data will be demonstrated mainly on the onboard TFT display, while PG13 and PG14 will be used only as activity or status indicators.

3. Workflow

3.1. Work with a teacher

The lecturer demonstrates the STM32F429I-DISC1 board, explains the I2C bus wiring principle, shared ground, 7-bit addressing and the polling communication sequence, and then students start to program individually.

3.1.1. STM32CubeIDE

The detailed project configuration has already been discussed in previous laboratory works, therefore here only the essential configuration windows for laboratory work No. 6 are important. In the adapted STM32F429I-DISC1 project the display-related peripherals may already be prepared, while students must configure I2C3, its PA8 and PC9 GPIO pins, and the application logic required by their own variant. In all variants use the same 7-bit I2C slave address 0x3C. In contrast to SPI, for I2C it is necessary to select the I2C peripheral mode, open-drain pin configuration and valid bus timing.

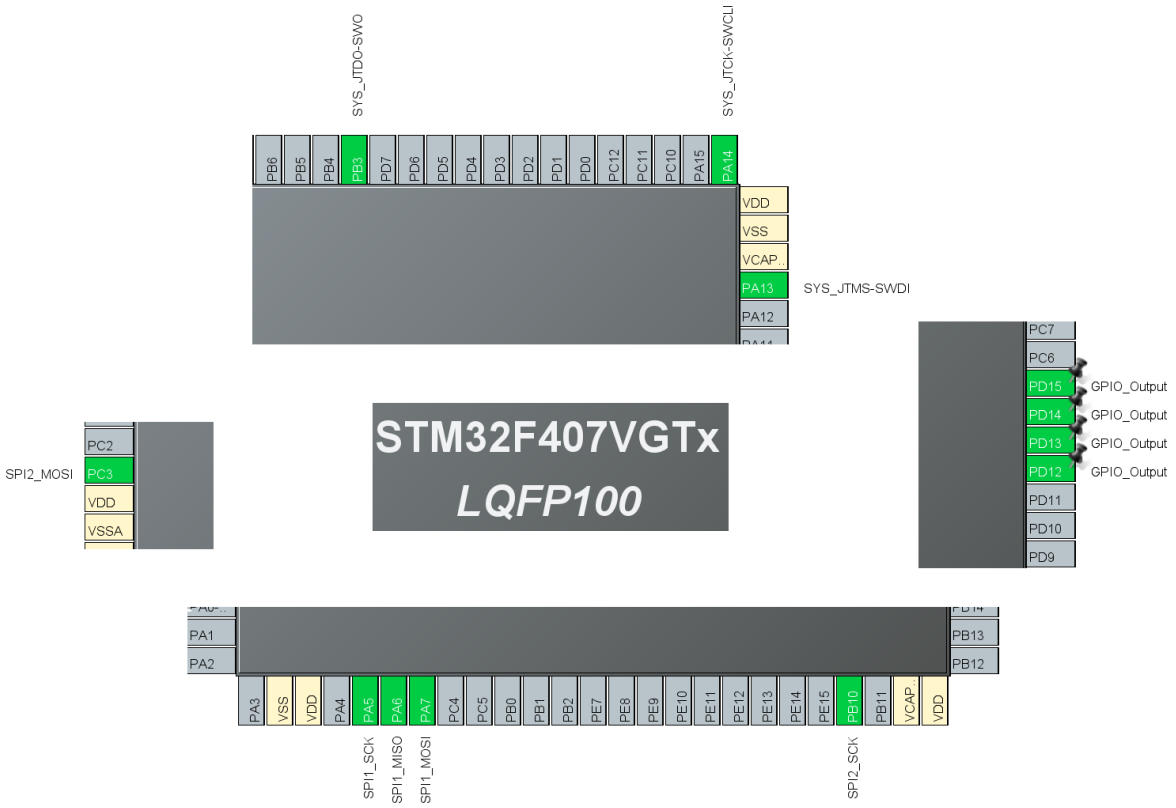


Fig. 1. Example of the STM32CubeMX pinout view for the adapted STM32F429I-DISC1 project. In LAB6 the important communication pins are I2C3_SCL on PA8 and I2C3_SDA on PC9.

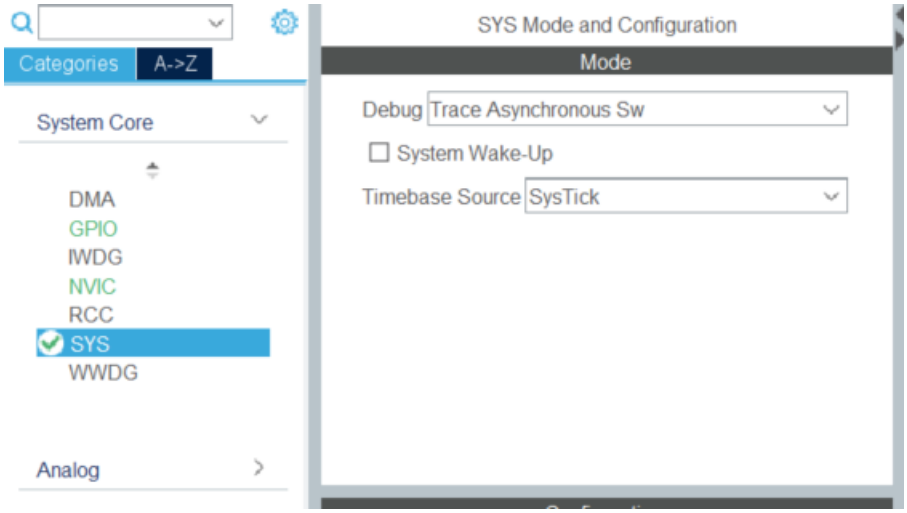


Fig. 2. Example of the STM32CubeMX workflow when reviewing pin assignment and peripheral use. For LAB6 verify that the board resources needed for the TFT display remain enabled and that PA8 and PC9 are assigned to I2C3.

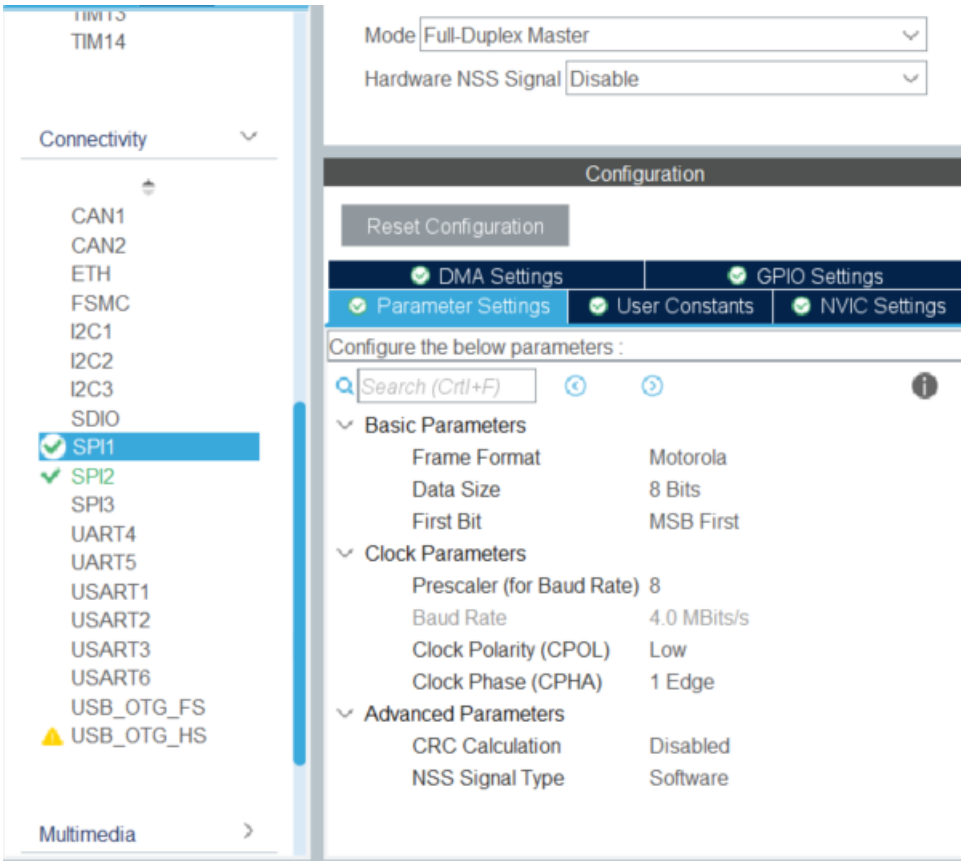


Fig. 3. Example of a peripheral parameter window in STM32CubeMX. For master variants configure I2C3 timing for address 0x3C and prepare the fixed transmit values: byte 0x2A, sequence {0x11, 0x22, 0x33}, cyclic sequence {0x10, 0x20, 0x30, 0x40}, or sequence {0x44, 0x55, 0x66} depending on the variant.

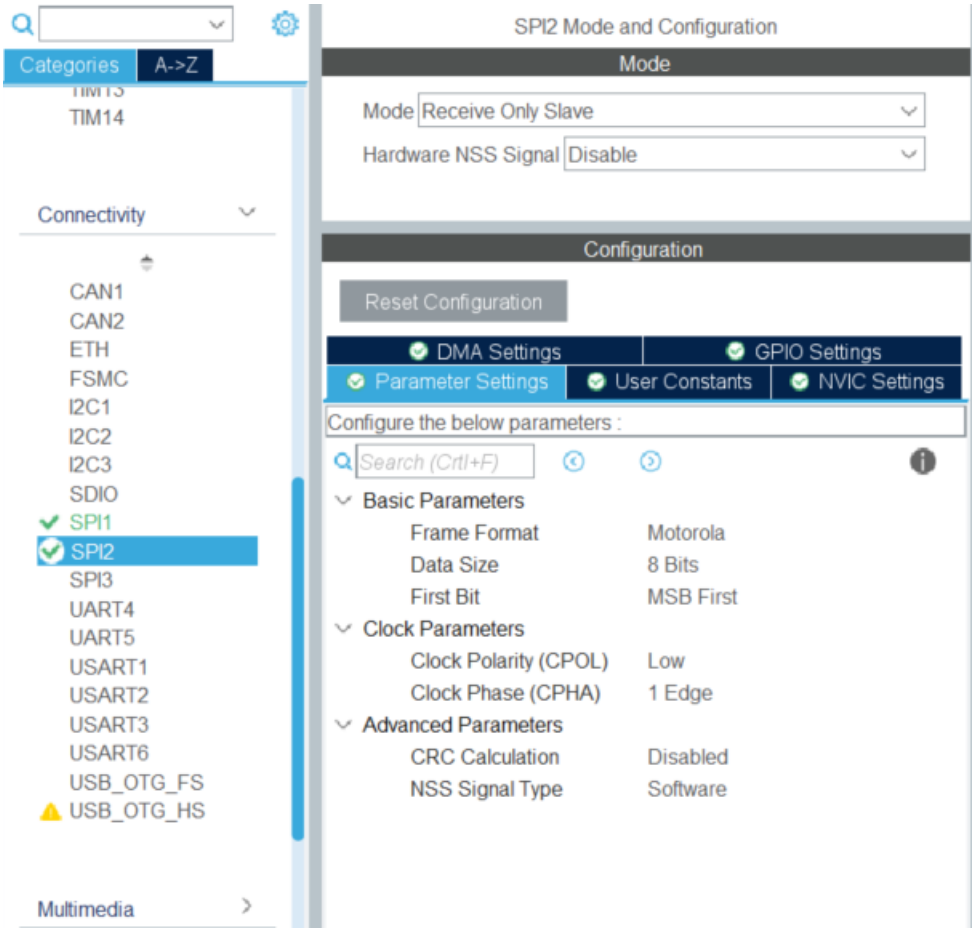


Fig. 4. Example of the final configuration review before code generation. For slave variants verify the I2C3 own address, the polling receive logic and the TFT output text for byte 0x2A or for the selected fixed byte sequence.

Note. The figures are illustrative examples of CubeMX configuration windows. In every specific variant the active communication peripheral must be adapted to I2C3 and the TFT output must match the fixed LAB6 values: address 0x3C, byte 0x2A, sequence {0x11, 0x22, 0x33}, cyclic sequence {0x10, 0x20, 0x30, 0x40}, or sequence {0x44, 0x55, 0x66}.

3.1.2. Template of the code

Examples of code templates are provided in the descriptions of previous laboratory works. In this adapted version it is practical to work with main.c, stm32f4xx_it.c and a separate application file that initializes I2C3 communication, TFT output and simple LED status indication.

3.1.3. Main functions in laboratory work No. 6

- HAL_I2C_Master_Transmit(…) – blocking mode: transmit one byte or a three-byte frame from the master board in the mandatory and additional master variants.
- HAL_I2C_Slave_Receive(…) – blocking mode: receive one byte or a three-byte frame on the slave board in the mandatory and additional slave variants.
- HAL_Delay(…) – define the transfer period in the polling loop and make repeated transmission easier to observe during testing.
- HAL_I2C_GetError(…) – inspect I2C bus errors, NACK conditions or timeout situations while debugging unsuccessful transfers.
- HAL_GPIO_WritePin(…) – control PG13 and PG14 as transfer, success or error indicators.
- BSP_LCD_DisplayStringAt(…) / BSP_LCD_DisplayStringAtLine(…) – display transmitted or received bytes, counters and checksum values on the onboard TFT display.

Data structures:

- uint8_t tx_buffer[3] = {0x11, 0x22, 0x33}; uint8_t tx_buffer_alt[3] = {0x44, 0x55, 0x66}; uint8_t tx_cycle[4] = {0x10, 0x20, 0x30, 0x40}; – fixed arrays used in the additional and cyclic master variants.

- uint8_t rx_byte = 0; uint8_t rx_buffer[3] = {0}; – variables used to store the received byte or three-byte I2C sequence.

3.2. Work by individual task

Students independently perform the assigned individual variants V6.1-V6.4. They must connect the required jumper wires between two STM32F429I-DISC1 boards, run their own I2C3 master or slave configuration and demonstrate on the TFT screen the fixed LAB6 values: address 0x3C, byte 0x2A, sequence {0x11, 0x22, 0x33}, cyclic sequence {0x10, 0x20, 0x30, 0x40}, or sequence {0x44, 0x55, 0x66}, depending on the chosen variant. When faced with difficulties, students can ask specific questions to the teacher.

3.3. Literature

1. Slides of the lectures in the moodle.vgtu.lt system.
2. STM32CubeIDE / STM32CubeMX code examples and the provided LAB6 project template.
3. STM32F429xx reference manual and datasheet.
4. STM32F429I-DISC1 user manual.
5. STM32F429I-DISC1 board schematic and pinout information.
6. I2C examples from the STM32CubeF4 firmware package and HAL documentation.
7. Materials from previous laboratory works and the adapted LAB6 code example.